# Exercise # 1.3

**Question 16 . Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent**

Take L.H.S $(p \leftrightarrow q)$

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ Bi – Conditional law

$\equiv (p \rightarrow q) \wedge (q \rightarrow p)$ Conditional Statement $p \rightarrow q \equiv \neg p \vee q$

$\equiv (\neg p \vee q) \wedge (\neg q \vee p)$ Distributive Law $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

$\equiv (\neg p \wedge \neg q) \vee (\neg p \wedge p) \vee (q \wedge \neg q) \vee (q \wedge p)$ Negation Law $p \wedge \neg p \equiv F$

$\equiv (\neg p \wedge \neg q) \vee F \vee F \vee (q \wedge p)$ Identity laws $p \vee F \equiv p$

$\equiv (\neg p \wedge \neg q) \vee (q \wedge p)$

. Commutative law $p \wedge q \equiv q \wedge p$

$\equiv (\neg p \wedge \neg q) \vee (p \wedge q)$

Question 17. Show that $\neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ are logically equivalent

Take L.H.S $\neg(p \leftrightarrow q)$

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ **Bi – Conditional law**

$\equiv (p \rightarrow q) \wedge (q \rightarrow p)$ **Conditional Statement** $p \rightarrow q \equiv \neg p \vee q$

$p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$

$\neg(p \leftrightarrow q) \equiv \neg \{ (\neg p \vee q) \wedge (\neg q \vee p) \}$

$\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$

**Now take R.H.S** $p \leftrightarrow \neg q$

$p \leftrightarrow \neg q \equiv (p \rightarrow \neg q) \wedge (\neg q \rightarrow p)$ **Bi – Conditional law**

$\equiv (p \rightarrow \neg q) \wedge (\neg q \rightarrow p)$ **Conditional Statement** $p \rightarrow \neg q \equiv \neg p \vee \neg q$

**Question 18 . Show that p → q and ¬q → ¬p are logically equivalent.**

**Take L.H.S p → q**

$p \rightarrow q$ , **Conditional Equivalent** $p \rightarrow q \equiv \neg p \vee q$

$p \rightarrow q \equiv \neg p \vee q$

**Now take R.H.S** $\neg q \rightarrow \neg p$

$\neg q \rightarrow \neg p$ , **Conditional Equivalent** $q \rightarrow p \equiv \neg q \vee p$

$\equiv \neg(\neg q) \vee \neg p$

$\equiv q \vee \neg p$ , **Commutative laws** $p \vee q \equiv q \vee p$

$\neg q \rightarrow \neg p \equiv \neg p \vee q$

**Question 19** Show that $\neg p \leftrightarrow q$ and $p \leftrightarrow \neg q$ are logically equivalent

Take L H S $\neg p \leftrightarrow q$

$\neg p \leftrightarrow q$ , Bi – Conditional law $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

$\equiv (\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$

, Conditional Statement $p \rightarrow q \equiv \neg p \vee q$

$\equiv (\neg(\neg p) \vee q) \wedge (\neg q \vee \neg p)$

$\equiv (p \vee q) \wedge (\neg q \vee \neg p)$

Now take R.H.S $p \leftrightarrow \neg q$

$p \leftrightarrow \neg q$ , Bi – Conditional law $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

$\equiv (p \rightarrow \neg q) \wedge (\neg q \rightarrow p)$

, Conditional Statement $p \rightarrow q \equiv \neg p \vee q$

$\equiv (\neg p \vee \neg q) \wedge (\neg(\neg q) \vee p)$

$\equiv (\neg p \vee \neg q) \wedge (q \vee p)$ , Commutative laws $p \vee q \equiv q \vee p$

$p \leftrightarrow \neg q \equiv (p \vee q) \wedge (\neg q \vee \neg p)$

**Question 20** **Show that $\neg(p \oplus q)$ and $p \leftrightarrow q$ are logically equivalent .**

**Take L.H.S $\neg(p \oplus q)$**

$\neg( p \oplus q)$ , **Exclusive Law** $p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

$p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

$\neg( p \oplus q) \equiv \neg \{(p \wedge \neg q) \vee (\neg p \wedge q) \}$

$\equiv (\neg p \vee q ) \wedge ( p \vee \neg q )$

**, Bi – Conditional law** $p \leftrightarrow q \equiv (\neg p \vee q ) \wedge ( p \vee \neg q )$

$\neg( p \oplus q) \equiv p \leftrightarrow q$

**Question 25. Show that $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent.**

**Take L.H.S $(p \rightarrow r) \vee (q \rightarrow r)$**

$(p \rightarrow r) \vee (q \rightarrow r)$ , **Conditional Statement** $p \rightarrow q \equiv \neg p \vee q$

$\equiv (\neg p \vee r) \vee (\neg q \vee r)$

**Distributive laws**

$\equiv (\neg p \vee \neg q) \vee r$

**Take R.H.S $(p \wedge q) \rightarrow r$**

$(p \wedge q) \rightarrow r$ , **Conditional Statement** $p \rightarrow q \equiv \neg p \vee q$

$\equiv \neg(P \wedge q) \vee r$

$\equiv (\neg p \vee \neg q) \vee r$

**Question 26. Show that ¬p → (q → r) and q → (p V r) are logically equivalent.**

Take L.H.S ¬p → (q → r)

(q → r) , Conditional Statement $p \rightarrow q \equiv \neg p \vee q$

≡ ¬p → (¬q V r)

, Conditional Statement $p \rightarrow q \equiv \neg p \vee q$

≡ p V (¬q V r )

Now Take R.H.S q → (p V r)

, Conditional Statement $p \rightarrow q \equiv \neg p \vee q$

≡ ¬q V ( p V r )

, Associative laws $(p \vee q) \vee r \equiv p \vee (q \vee r)$

≡ p V (¬q V r )

**Question 34. Find the dual of each of these compound propositions.**

**a) $p \vee \neg q$**

**b) $p \wedge (q \vee (r \wedge T))$**

**c) $(p \wedge \neg q) \vee (q \wedge F)$**

**Question 35. Find the dual of each of these compound propositions.**

**a) $p \wedge \neg q \wedge \neg r$**

**b) $(p \wedge q \wedge r) \vee s$**

**c) $(p \vee F) \wedge (q \vee T$**